

3 Plants have specialised cells for the efficient transport of assimilates.

(a) Table 3.1 shows some of the features of two different types of cell found in plant tissue, which are adapted for the efficient transport of assimilates.

Table 3.1

feature of cell	cell type A	cell type B
cytoplasm	✓	✓
nucleus	x	✓
mitochondria	few	many
cellulose in the cell wall	✓	✓
ribosomes	x	✓

Key
✓ present
x absent

Many plasmodesmata connect type **A** cells with type **B** cells.

Identify cell type **A** and explain why the plasmodesmata are important.

cell type **A**

explanation

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- (d) Water that has travelled through xylem vessels reaches the leaves. Cooling of the leaf occurs as a result of the evaporation of water during transpiration.

Water has a high latent heat of vaporisation because water molecules form hydrogen bonds.

With reference to hydrogen bonding, suggest why cooling of the leaf occurs as a result of evaporation of water during transpiration.

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